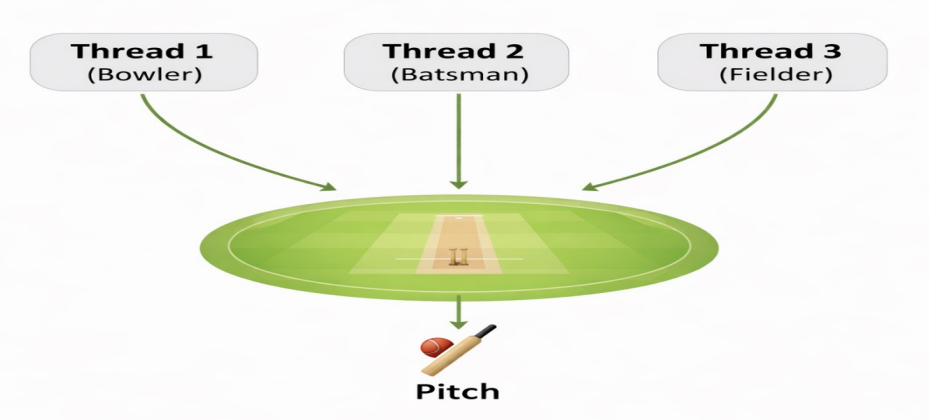
T20 Cricket Simulator

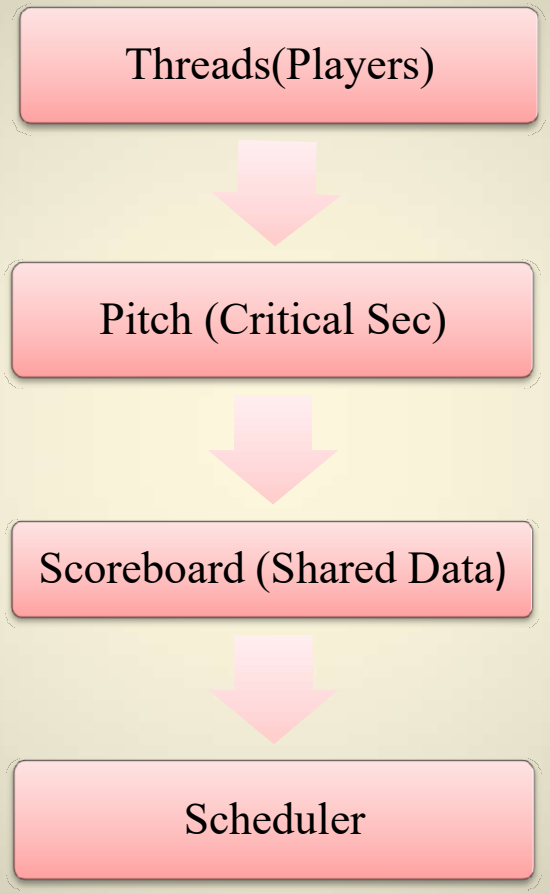
Operating Systems Project (CSC-204)

Problem Statement

Aspect	Description
• Goal	• Simulate T20 Match
• Approach	• Multithreading
• Challenge	• Synchronization
• Risk	• Deadlocks



System Architecture



Thread Model

Thread Type	Count	Role
Bowler	1	Writes to pitch
Batsman	2	Reads + updates score
Fielders	10	Passive / signal-based

Synchronization

Feature	Without Sync	With Sync
Data Safety	Corrupted	Safe
Multiple Access	Yes	Controlled
Race Condition	Yes	No
Dirty Reads	Yes	No

Critical Section

Pitch Access Control

- A critical section is a part of the program where **shared resources are accessed**
- In our project, the **Pitch acts as the critical section**

➢ **Working Mechanism (Step-wise)**

1. Thread requests access to pitch
2. Mutex lock is checked
3. If free → thread enters
4. Executes (ball delivery / shot)
5. Releases lock after completion

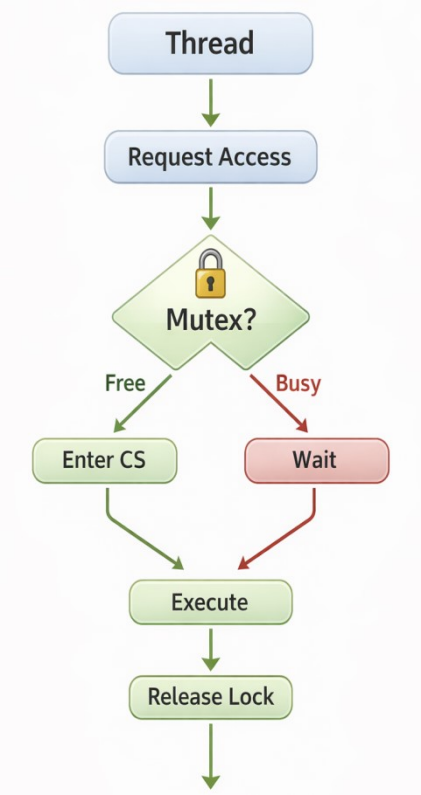
Component	Role
Pitch	Shared Resource
Bowler	Writes (delivers ball)
Batsman	Reads & update score
Mutex	Controls access

➢ **Without Critical Section**

Without Control	With Mutex
Multiple bowlers at once	Only one bowler
Data corruption	Safe data
Race condition	No race
Wrong score	Accurate score

Example Scenario:

Two bowlers try to bowl at same time
Without mutex → collision of data
With mutex → one waits, one executes



Only one thread can access the pitch at a time to ensure consistency.

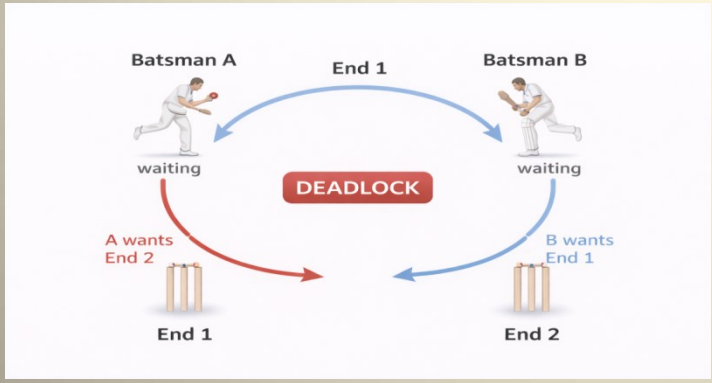
Scheduling

Scheduling Type	Used?	Reason
FCFS	✕	Not fair
SJF	✕	Not realistic
Round Robin	✔	Fair + real match

Deadlock Handling

Condition	Present?
Mutual Exclusion	✔
Hold & Wait	✔
No Preemption	✔
Circular Wait	✔

Result: **Deadlock Occurs**



Solution:

Umpire (kernel)

↓

Detect deadlock

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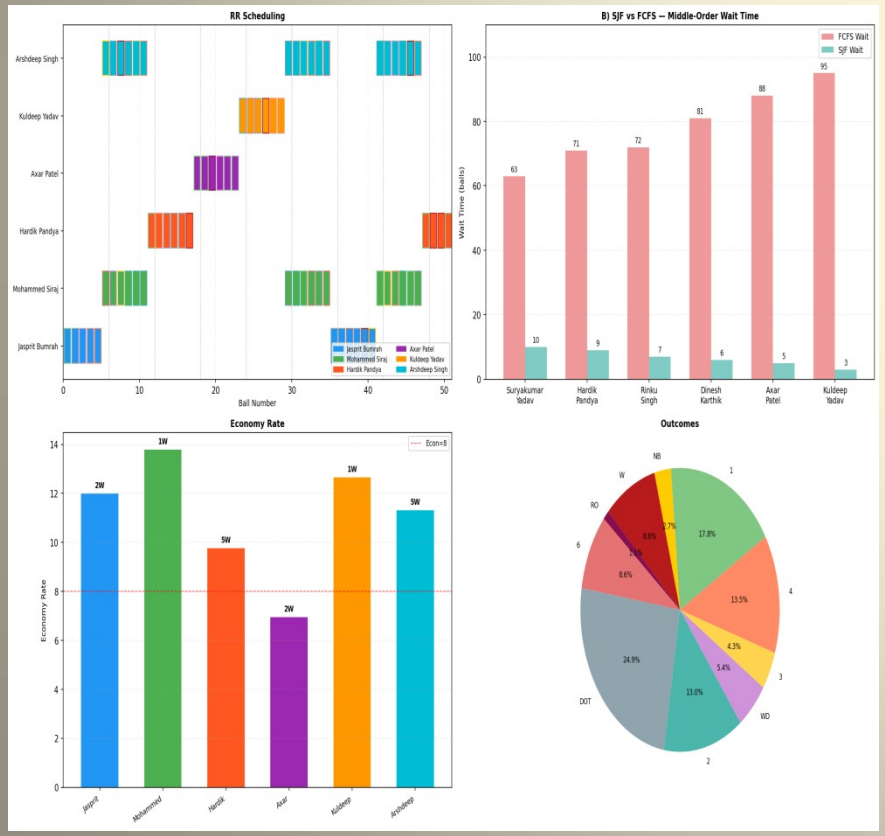
Terminate one process

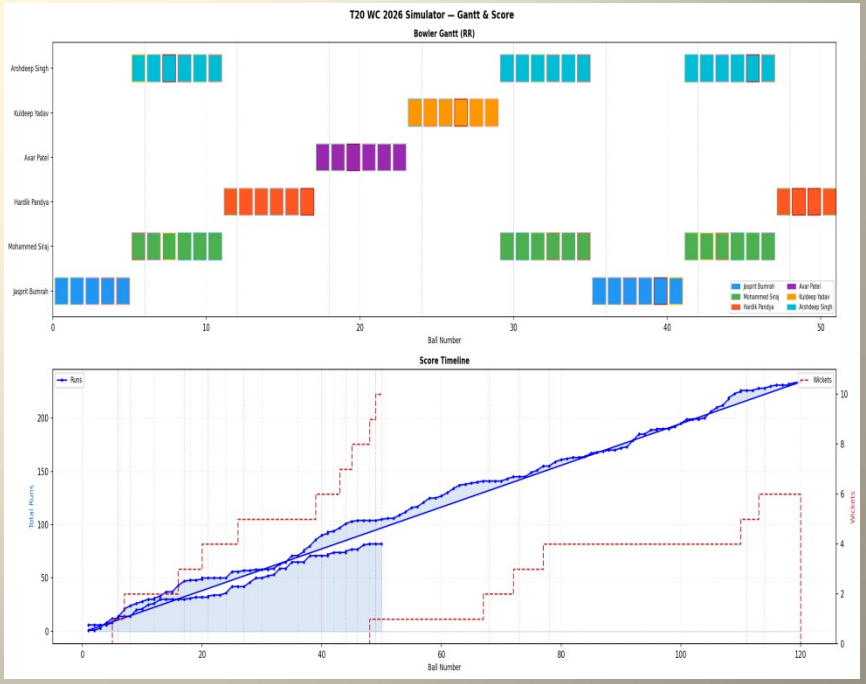
Outputs & Results

Content:

- Ball-by-ball match log generated
- Gantt chart shows thread execution
- Analysis charts show performance

Output	Description
Match Log	Ball-by-ball events
Gantt Chart	Thread scheduling
Analysis	Performance comparison





Analysis & Comparison

Scheduling	Performance
FCFS	High waiting time
SJF	Not practical
Round Robin	Balanced & fair

“Round Robin gives best balance for simulation”

Conclusion + Assumptions

➢**Conclusion:**

•OS concepts successfully implemented

•Efficient multithreading achieved

•Deadlock handled properly

➢**Assumptions:**

•Simplified cricket rules

•Fixed number of players

•Controlled environment

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